



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Allugundu Mohammad Naheed

Reg. No:192525336

Course Code: CSA0509

Slot: A

Course Name: Database Management System

Course Faculty: Dr. R Kesavan

Project Title: Design and Implementation of a Secure Electronic Voting Machine with Voter Verification System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

The first photograph shows a student standing next to a presentation screen displaying the project title. The second photograph is a screenshot of the 'Smart Water Utility Management System' dashboard, which includes a 'Utility Dashboard' with metrics like Connections (2,000,000), Today's Consumption (48.6 ML), Pending Bills (₹84,250), Leak Alerts (17), and Open Complaints (24). The third photograph is a screenshot of the 'Payment Processing' and 'Transaction Log' interface, showing a table with Transaction ID, Operation, and Status.

Project Description:

The **Smart Water Utility Consumption, Billing, and Complaint Management System** is designed to manage large-scale water utility data efficiently, including customer connections, meter readings, monthly billing, payments, and complaints. The system addresses the problems of sequential file organization by using structured database tables and efficient data-access methods. It allows utility staff to quickly retrieve the latest consumption readings, view historical usage, generate bills, record payments, and track complaints.

The system focuses on important DBMS concepts such as **file organization, B/B+ tree indexing, hashing, query processing, and query optimization**. Indexes are used to reduce the need for full-table scans and improve searches based on connection IDs, timestamps, billing cycles, and complaint status. SQL operations such as **JOIN, GROUP BY, aggregate functions, filtering, and subqueries** are used to generate useful consumption, billing, and operational reports efficiently.

Transaction management is another major component of the system. **ACID properties, COMMIT, ROLLBACK, SAVEPOINT, and transaction isolation** are used to ensure that billing and payment operations remain accurate even when multiple users access the database simultaneously. Duplicate bills, lost payments, and overwritten complaints are prevented through constraints and controlled transactions. Overall, the system provides a scalable and reliable database solution for modern smart-water utility management.

Student Signature

Faculty Signature